

The Regulator

By Tim Surtell and John Hewes

While there are many circuits that will tolerate a smoothed power supply, some must have a completely regular supply with no ripple voltage. This article discusses regulator ICs which can provide this regular power supply.

The 78xx Series of Regulators

There are many types of regulator IC and each type will have different pin-outs and will need to be connected up slightly differently. Therefore, this article will only look at one of the common ranges of regulator, the 78xx series.

There are seven regulators in the 78xx series, and each can pass up to 1A to any connected circuit. There are also regulators with similar type numbers that can pass a higher or lower current, as shown in the table below. In addition, variable regulators are available, as are regulators that can provide negative regulation voltages for circuits that require them.

Type Number	Regulation Voltage	Maximum Current	Minimum Input Voltage
78L05	+5V	0.1A	+7V
78L12	+12V	0.1A	+14.5V
78L15	+15V	0.1A	+17.5V
78M05	+5V	0.5A	+7V
78M12	+12V	0.5A	+14.5V
78M15	+15V	0.5A	+17.5V
7805	+5V	1A	+7V
7806	+6V	1A	+8V
7808	+8V	1A	+10.5V
7812	+12V	1A	+14.5V
7815	+15V	1A	+17.5V
7824	+24V	1A	+26V
78S05	+5V	2A	+8V
78S09	+9V	2A	+12V
78S12	+12V	2A	+15V
78S15	+15V	2A	+18V



- Datasheets**
- ▶ [78Lxx Series Regulator ICs](#) - PDF format (199kb)
 - ▶ [78xx Series Regulator ICs](#) - PDF format (148kb)
 - ▶ [78Mxx Series Regulator ICs](#) - PDF format (311kb)

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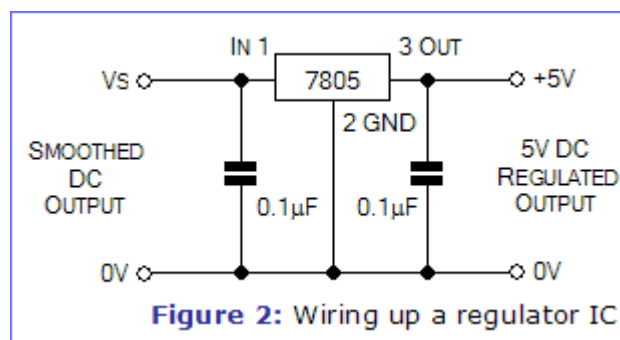
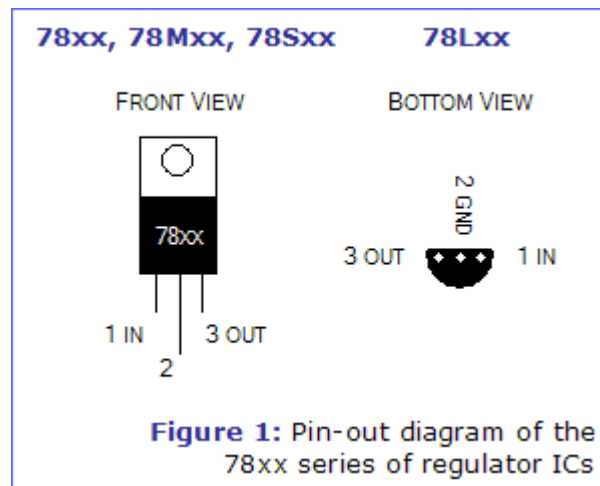
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Ripple Voltage

If you are using a regulator after the smoothing block of the power supply, then you shouldn't need to worry about the ripple voltage, since the whole point of using a regulator is to get a stable, accurate, known voltage for your circuits! However, if the ripple voltage is too large and the input voltage to the regulator falls below the regulated voltage of the regulator, then of course the regulator will not be able to produce the correct regulated voltage. In fact, the input voltage to a regulator should usually be at least 2V above the regulated voltage. In our power supply circuit, the input to the 7805 regulator is around 12V, and the regulation voltage is 5V, so there is plenty of headroom. The maximum input voltage to any 78xx regulator is 30V.

78xx Pin-out

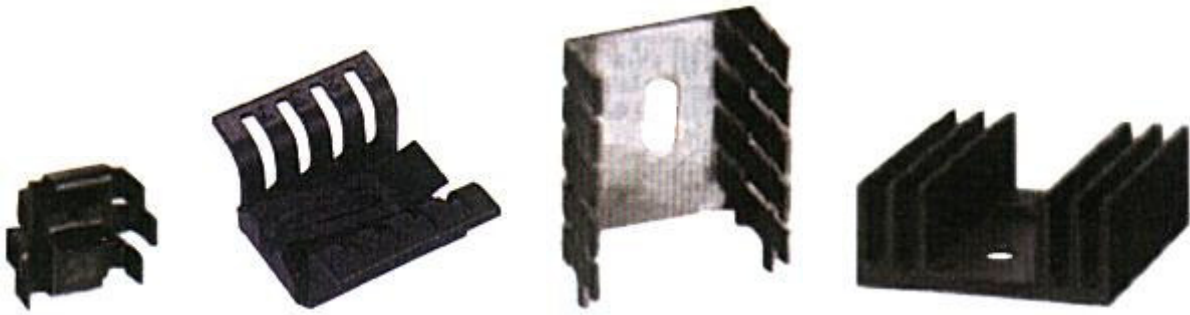
The 78xx, 78Mxx, and 78Sxx regulators all have the pin-out shown in the left of figure 1 and are normally supplied in a case style known as TO-220. The 78Lxx series, shown in the right of figure 1, also has the same pin-out but has a case style known as TO-92. They are all connected to the rest of the power supply in the same way, as shown in figure 2.



Heatsinks

In use, a regulator IC will get quite hot, so a heatsink will need to be attached to it to dissipate the heat. The type of heatsink you choose depends on the regulator's case style, the amount of heat it must dissipate, and the way in which you wish to mount it.

The examples shown below all suit the 78xx, 78Mxx, and 78Sxx regulator series TO-220 case style. The first two are the simplest and are clip-on types. The third has a hole so it can be bolted to the regulator, and two legs that can be soldered to a piece of stripboard or PCB. The fourth again has a mounting hole for the regulator, and also has a mounting hole for fixing to a board. To be sure of good heat transfer from the regulator to the heatsink, you can sandwich heat transfer compound between them.



Choosing a Heatsink

Heatsinks are rated by their 'thermal resistance' (R_{th}) given in $^{\circ}\text{C}/\text{W}$. For example, a rating of $2^{\circ}\text{C}/\text{W}$ means the heatsink (and therefore the regulator attached to it) will be 2°C hotter than the surrounding air for every 1W of heat it is dissipating. Note that a lower thermal resistance means a better heatsink.

To determine the heatsink rating required, follow these steps:

1. **Work out the thermal power to be dissipated (P)**

To do this, use the following formula...

$$P = I(V_s - V_r)$$

...where P is the power to be dissipated, V_s is the input voltage to the regulator, V_r is the regulated voltage, and I is the current being supplied to the connected circuit. For example, a 5V regulator supplying 0.5A and powered by 12V supply, would dissipate...

$$P = 0.5 \times (12 - 5) = 3.5\text{W}$$

Note the use of $(12 - 5) = 7\text{V}$ which is the voltage difference across the regulator. Clearly, lower supply voltages are better so long as they are a few volts above the regulated output.

2. **Find out the maximum operating temperature of the regulator (T_{max})**

This can be found from a catalogue or datasheet. For a typical 78xx series regulator, $T_{max} = 125^{\circ}\text{C}$.

3. **Estimate the maximum ambient (surrounding air) temperature (T_{air})**

If heatsink is not in a case, $T_{air} = 30^{\circ}\text{C}$ is reasonable, but inside a case it will be higher (perhaps 40 or 50°C) allowing for everything to warm up in operation.

4. **Work out the maximum thermal resistance for heatsink (R_{th})**

Use the following formula...

$$R_{th} = \frac{(T_{max} - T_{air})}{P}$$

...where R_{th} is the thermal resistance, T_{max} is the operating temperature from (2) above, T_{air} is the ambient temperature from (3) above, and P is the power to be dissipated from (1) above.

So with the example figures given above...

$$R_{th} = \frac{(125 - 30)}{3.5} = 27^{\circ}\text{C}$$

5. Choose a heatsink with a thermal resistance which is LESS than the value calculated above (remember a lower value means better heatsinking!) In this case, 10°C/W would be a sensible choice to allow a safety margin. A 10°C/W heatsink dissipating 3.5W will have a temperature difference of $10 \times 3.5 = 35^{\circ}\text{C}$ so the temperature of the regulator will rise to $30^{\circ}\text{C} + 35^{\circ}\text{C} = 65^{\circ}\text{C}$ which is safely less than the 125°C maximum.

The heatsink shown on the far right above can dissipate 9.5°C/W and so would be ideal for use in this example.



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7812 1A Regulator	1	CH36P	£0.59	£0.59
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78M05 0.5A Regulator	1	QL28F	£0.69	£0.69
78M12 0.5A Regulator	1	QL29G	£0.69	£0.69
78M15 0.5A Regulator	1	QL30H	£0.69	£0.69
78S05 2A Regulator	1	UJ54J	£0.77	£0.77
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78S15 2A Regulator	1	UJ57M	£0.77	£0.77
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